

Character class:

- This is the base class for this term project
- The character class's member variables contain:
- Character Name
- Level
- Health Points (HP)
- Attack Power
- Defense
- Experience Points
- Status (Alive / Defeated)
- It's member functions are:
 - `Character()`; - the constructor
 - `setName(const std::string n)` which is a void function for storing the name
 - `std::string getName() const`; - this is a getter function for name
 - `void setCharacterType(const std::string &ct)`; - this is a setter function for charactertype
 - `std::string getCharacterType() const`; - this is a getter function for charactertype
 - `void setLevel(int l)`; - this is a setter function for level
 - `int getLevel() const`; - this is a getter function for level
 - `void setHealth(int h)`; - this is a setter function for health
 - `int getHealth() const`; - this is a getter function for health
 - `void setAttackPower(int ap)`; - this is a setter function for attack power
 - `int getAttackPower() const`; - this is a getter function for attack power
 - `void setDefense(int d)`; - this is a setter function for defense
 - `int getDefense() const`; - this is a getter function for defense
 - `void setExperiencePoints(int xp)`; - this sets the experience points for the characters
 - `int getExperiencePoints() const`; - this is a getter function for experience points.
 - `void setStatus(const std::string &s)`; - this sets the status of the character
 - `std::string getStatus() const`; - this function gets the status if it's alive or defeated
 - **virtual void specialSkill(Character &target);** // Pure virtual function for special skill – This will allow other classes to override its own specialSkill
 - `~Character()`; - The destructor

The following classes are inherited from Character:

Knight:

- The member variables are shield and sword which are part of a separate class called Weapon.
- The member functions are:
- `Knight()`; - constructor for Knight
- `void swordAttack(Character &target)`; - this is a void function for attacking with sword
- `void shieldBlock(Character &target)`; - this is a void function for blocking with shield
- `void shieldBash(Character &target)`; - this is a void function for attacking with shield
- `void specialSkill(Character &target) override`; - this is overridden from the character class
- `~Knight()`; - destructor for Knight

Mage:

- The member functions are:
- `Mage()`; - constructor for Mage

- **void** Fireball(Character &target); - This is a void function for when the Mage casts a Fireball
- **void** IceShard(Character &target); - This is a void function for when the Mage casts a IceShard
- **void** lightningStrike(Character &target); - This is a void function for when the Mage casts a lightning strike
- **void** specialSkill(Character &target) **override**; //function overridden from character
- ~Mage(); - destructor for mage

Doctor:

- The member functions are:
- Doctor(); - constructor for doctor
- **int** Heal(Character &patient); - int function for healing character
- **void** specialSkill(Character &patient) **override**; - overridden from Character class
- ~Doctor(); - destructor for doctor

Weapon:

- This is a separate class for Knight
- Its member variables are weaponName and damage which both stores the name of the weapon and the damage.
- Its member functions are:
- Weapon(); - constructor for Weapon
- Weapon(string wn, **int** d); - parameterized constructor for Weapon
- **void** setWeaponName(string wn); - setter function for weapon name
- string getWeaponName(); - getter function for weapon name
- **void** setDamage(**int** d); - setter function for damage
- **int** getDamage(); - getter function for damage
- ~Weapon(); - destructor for Weapon